

Aggregate Functions in SQL

Q. What are Aggregate Functions in SQL Explain the functions COUNT(), SUM(), AVG(), MAX(), and MIN() with syntax and appropriate examples.

[DEF] Definition to Remember

Aggregate Functions in SQL operate on a set of values (an entire column) and return a single summarizing value. They are used for data analysis and reporting, typically combined with the GROUP BY clause. With the exception of COUNT(*), all aggregate functions ignore NULL values.

The Five Standard Aggregate Functions

Function	Returns	Works On
COUNT	Number of rows	Any type
SUM	Total sum	Numeric only
AVG	Arithmetic mean	Numeric only
MAX	Highest value	Numeric, Text, Date
MIN	Lowest value	Numeric, Text, Date

1. COUNT()

Return

2. SUM()

Calcula

3. AVG()

Calcula

4. MAX()

Return

5. MIN()

Return

NULL Behavior in Aggregate Functions

* SUM(), AVG(), MAX(), MIN() -- ignore NULL values.

* CO

[MUST-WRITE] Must-Write Points (for 10 marks)

- 1. Aggregate functions compute a single value from a set of values (a column).
- 2. COUNT(*) counts all rows including NULLs; COUNT(column) excludes NULLs.

3. SUM() and AVG() work on numeric columns only.
4. MAX() and MIN() work on numeric, text (alphabetical), and date values.
5. All aggregate functions ignore NULL values -- except COUNT(*).
6. Use DISTINCT inside aggregate functions to calculate on unique values only.
7. Aggregate functions are most powerful when combined with GROUP BY and HAVING.

[RECALL] Quick Recall

Aggregate Functions -> COUNT(*) / COUNT(col) -> SUM (numeric) -> AVG (numeric) -> MAX / MIN (any type) -> Ignore NULLs (except COUNT*) -> Used with GROUP BY